

Operational Best Practices & Equipment Usage in Mining

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Mining efficiency and safety depend on adherence to operational protocols and equipment standards. Key practices include:

1. ****Drilling and Blasting Procedures****

All blasting must follow controlled timing and explosive charge protocols. Seismographic monitoring is used to limit overbreak and vibration.

2. ****Equipment Maintenance****

Preventive maintenance schedules for trucks, drills, crushers, and conveyors are mandatory. Downtime logs and performance metrics must be tracked.

3. ****Ore Handling and Transportation****

Conveyance systems must be regularly inspected. Trucks must be loaded and routed to minimize fuel use and maximize haul cycle efficiency.

4. ****Shift Handover Protocol****

A standardized log system for operator shift changes is enforced to ensure continuity and awareness of equipment status and operational issues.

5. ****Technology Integration****

Use of telematics, drones, and automated systems must follow IT and cybersecurity protocols. Training is mandatory before use of autonomous equipment.

6. ****Contractor Management****

All third-party operators must comply with site-specific induction, safety protocols, and performance standards.

Effective operations are monitored through KPIs, real-time dashboards, and supervisory field checks to maintain productivity and compliance.